

Preparation list for Preliminary Exam (9744/4/2018)

Question 1

	Apparatus/Reagents/Chemicals	Quantity per candidate
1	M Each candidate must be provided with the following apparatus and materials. 60 cm³ of 15% UHT skimmed milk, labelled M. To make up 200 cm³, mix 30 cm UHT skimmed milk with 170 cm³ distilled water. This is sufficient for three candidates. The solution should be provided to candidates in 250 cm³ beakers labelled M. Depending on the activity of the protease in E1, E2 and E3, it may be necessary to modify the concentration of the UHT skimmed milk in M. To determine the activity of the protease, put 5cm³ of M into a boiling tube. Put 1 cm³ of E3 into this boiling tube, gently swirl and place in a water-bath at a temperature of between 35 °C and 40 °C. Immediately start a stopwatch. M should become significantly clearer within five minutes but should not reach the end-point (no further clearing) within the first two minutes. If M reaches the end-point within two minutes, increase the concentration of UHT skimmed milk in M and retest. If M does not clear significantly within five minutes, decrease the concentration of UHT skimmed milk in M and retest. Candidates should not be informed of any change in M.	1
2	E1 15 cm³ of 0.25% bacterial protease solution, labelled E1. To make up 100 cm³, sprinkle 0.25 g of bacterial protease 'powder-onto the surface of 80cm³ of distilled water, stirring continuously, until the protease is dissolved. Make up to 100cm³ with distilled water. This is sufficient for six candidates. The solution should be provided to candidates in plastic vials (B197) labelled E1, at room temperature.	1
3	E2 15 cm³ of 0.5% bacterial protease solution, labelled E2.	1

	To make up 100 cm³, sprinkle 0.5 g of bacterial protease powder onto the surface of 80 cm³ of distilled water, stirring continuously, until the protease is dissolved. Make up to 100 cm³ with distilled water. This is sufficient for six candidates. The solution should be provided to candidates in plastic vials (B197) labelled E2, at room temperature.	
4	E3 [HH] 15 cm³ of 1% bacterial protease solution, labelled E3. To make up 100 cm³, sprinkle 1 g of bacterial protease powder onto the surface of 80 cm³ of distilled water, stirring continuously, until the protease is dissolved. Make up to 100 cm³ with distilled water. This is sufficient for six candidates. The solution should be provided to candidates in plastic vials (B197) labelled E3, at room temperature. Bacterial protease powder and the 1% solution of bacterial protease (E3) are harmful [HH]. You are advised to wear eye protection and gloves, and wash off any spillages. The plastic vials should also be labelled with the safety hazard symbol stickers – health hazard, that have been provided for Centres' use.	1
5	30 cm ³ of distilled water in a beaker, labelled W .	1
6	500 cm ³ beaker to maintain a water-bath at a temperature of between 35 °C and 40 °C and able to hold five boiling tubes.	1
7	250 cm³ of water at about 50 °C in a beaker, labelled hot water. The Supervisor may use a thermostatically-controlled water-bath to provide additional hot water, if requested by candidates.	1
8	250 cm ³ of water at, or below, room temperature in a beaker, labelled cold water.	1
9	Thermometer (-10 °C to 110 °C)	1
10	Boiling tubes	5
11	Boiling tube rack(s) to hold five boiling tubes.	1
12	5 cm³ syringes If there is a shortage of syringes, candidates can be instructed to reuse them as long as the syringes are thoroughly washed with distilled water before reuse.	3

13	Stopwatch	1
14	250 cm ³ beaker with at least 200 cm ³ tap water, labelled washing water.	1
15	250 cm ³ beaker, labelled for waste.	1
16	Paper towels	5
17	Safety goggles	1 pair
18	Disposable gloves	1 pair
19	Permanent marker suitable for writing on glass	1

Question 2

Each candidate must have sole, uninterrupted use of a microscope and **B1** for 1hour 15 minutes only.

For each candidate:

- The microscope must be set up on the low-power objective lens.
- The slide **must not** be left on the stage of the microscope.

	Apparatus/Reagents/Chemicals	Quantity per candidate
1	Microscope with an eyepiece graticule	1 between 2
2	Prepared slide (blood smear), labelled B1	1 between 2
3	Stage micrometre	1 between 2